

cocotb ↔ UVM roles

Plain cocotb is Python driving a simulator with async triggers and DUT handles

Six role pairs

- Six pairs bridge the worlds
- Cocotb test decorator and Makefile module versus UVM test plus test name plusarg
- Time: await RisingEdge or Timer versus posedge and delay
- DUT access: dut handle versus virtual interface
- Drive: Python pin assigns versus driver through vif
- Structure: pyuvm agent versus uvm agent

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

cocotb vs UVM map

Map **cocotb** / **pyuv** roles onto **SV UVM**: tests, awaits, DUT handles, drivers, agents, and checks. Starter: **DUT handle** ↔ **virtual interface** already selected and mapped.

Starter example: dut . signal ↔ virtual interface already selected and marked mapped.

Load starter example

Challenges 0 / 22 cleared

Quiz: cocotb: Plain cocotb is best described as...

☐ an async Python testbench library (triggers + DUT handles)

☐ a SystemVerilog synthesis subset

☐ a GTKWave theme pack

☐ a Makefile-only lint tool

Show hint

Check

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Core ideas

cocotb

Async Python TB: triggers + DUT handles.

pyuv

UVM-shaped components written in Python.

SV UVM

Same roles in SystemVerilog classes.

map roles

Jobs match; APIs and packaging differ.

Browser lab starter

Real cocotb practice

- In the real cocotb track, restate the map on paper before you open a Makefile or UVM tree
- Pick three pairs, test, await
- Optional: open the lab's source sketch and read the comment header
- This module is role literacy, not choosing pyuvm versus plain cocotb for your project yet

Pitfalls to watch

- Do not treat the map as one-to-one syntax, APIs differ even when roles match
- A common miss is assuming pyuvm is required for every cocotb bench
- Another trap is mapping driver to raw Python assigns when you already built a pyuvm agent
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and name what dut maps to on the UVM side
- On paper, fill in test and await for both columns without looking
- When you are ready